MY GRAPH

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1. *Summary*

Among the q^2+q+1 points in the finite projective plane PG(2, q), with q odd, $q+1$ lie on any given conic. The tangents at these $q+1$ points intersect in pairs in $\binom{q+1}{2}$ *external* points which are the poles of $\binom{q+1}{2}$ *secants*. The remaining $\binom{q}{2}$ points are *internal* (lying on no tangent); their polars are the $\binom{q}{2}$ *nonsecants* (the 'external lines' of Hirschfeld [**19**, p. 166]). Each nonsecant contains $q+1$ points, $\frac{1}{2}(q+1)$ of which are external (because the two tangents from each account for all the $q+1$ tangents); the remaining $\frac{1}{2}(q+1)$ are internal. Each secant contains two points on the conic and $\frac{1}{2}(q-1)$ external points (because the two tangents from each account for the tangents at the remaining $q-1$ points on the conic); the remaining $\frac{1}{2}(q-1)$ are internal. Thus the $\binom{q}{2}$ internal points and $\binom{q}{2}$ nonsecants form a self-dual configuration $\binom{q}{2}_{\frac{1}{2}(q+1)}$; the $\binom{q+1}{2}$ external points and $\binom{q+1}{2}$ secants form a configuration $\binom{q+1}{2}_{\frac{1}{2}(q-1)}$. A self-dual configuration n_d (or n_dn_d) consists of n points and n lines, with d of the lines through each point and d of the points on each line [**18**, pp. 95, 97].

Another way to express the same idea arises when we regard the points of the configuration as the vertices of a *graph*, two of them being joined by an edge whenever these points are *conjugate* (each lying on the polar of the other). We thus obtain a graph of valency (or 'degree') $\frac{1}{2}(q+1)$ with $\binom{q}{2}$ vertices, and a graph of valency $\frac{1}{2}(q-1)$ with $\binom{q+1}{2}$ vertices, each having $\frac{3}{4}\binom{q+1}{3}$ edges. (A similar idea occurred to Erdős, Rényi, and Sós [**16**, p. 218]; but they made no distinction between internal and external points.)

The significant cases (when the valencies $\frac{1}{2}(q\pm 1)$ are small enough to be manageable but large enough to be interesting) are those where $q=5$ and $q=7$.

When $q=5$, the 10_3 is the Desargues configuration and the 15_2 is a set of five separate triangles [**11**, pp. 117, 119; **7**, p. 243]. The corresponding graphs are the Petersen graph (Fig. 1) and the same set of triangles.

We shall find in § 5 that, when $q=7$, the configurations 21_4 and 28_3 are related to 'polyhedra' in unitary 3-space. The corresponding graphs are the edge graph of the Heawood graph [**2**, pp. 11, 236] and the so-called Coxeter graph [**2**, p. 241].

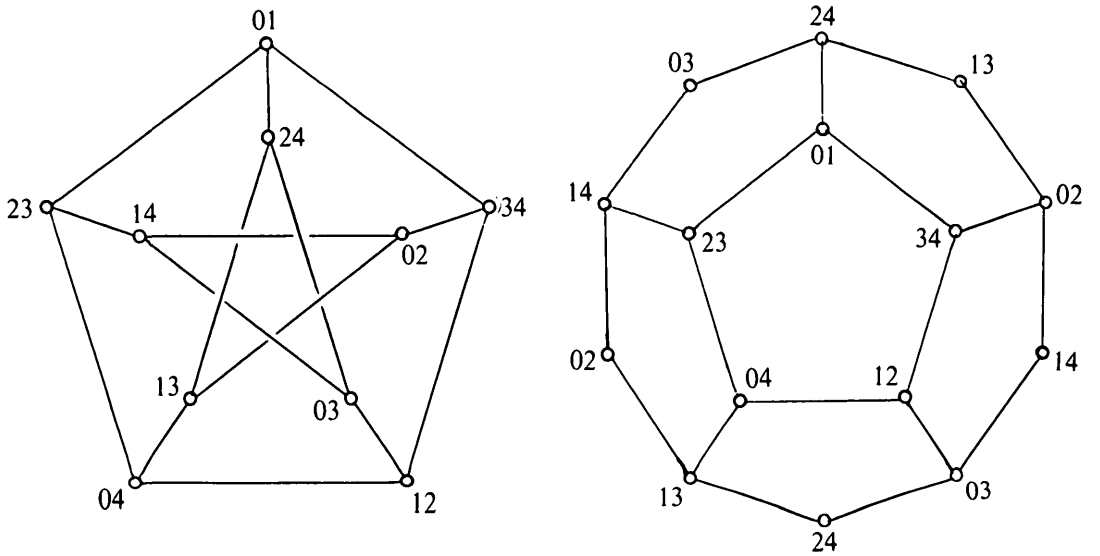


FIG. 1. The Petersen graph and the elliptic map $\{5,3\}_5$.

2. *Introduction*

Consider a trivalent graph whose 28 vertices are denoted by the $\binom{8}{2}$ unordered pairs of eight symbols consisting of ∞ and the residues

$$0,\ 1,\ 2,\ 3,\ 4,\ 5,\ 6 \pmod 7,$$

while its 42 edges are determined by the following two rules: *for every arithmetic progression* α, β, γ (mod 7), $\alpha\gamma$ *is joined to* $\beta\infty$; *and for every arithmetic progression* $\alpha, \beta, \gamma, \delta$ (mod 7), $\alpha\beta$ *is joined to* $\gamma\delta$.

These incidences are evidently preserved when we add 1 (or any other residue) to all the residues involved (including $6+1=0$ and $\infty+1=\infty$) and when we multiply them by any non-zero residue (including -1, which is 6). By an apparent miracle, the incidences are preserved also by reciprocation (replacing ∞, 0, 1, 2, 3, 4, 5, 6 by 0, ∞, 1, 4, 5, 2, 3, 6). For instance, the vertex 01 is joined to 23, 56, 4∞, and the 'reciprocal' vertex 1∞ is joined to 45, 36, 02. We see in this manner that the automorphism group of the graph is (or at least includes) PGL(2, 7), the group of linear fractional transformations of determinant ± 1 in the field GF[7], of order 336 [**10**, pp. 85, 87].

By drawing the graph as in Fig. 2, with three heptagons joined to seven 'extra' vertices $\beta\infty$, we recognize it to be the one which Tutte [**25**] ascribed to me. Tutte proved that it is, like the Petersen graph, both non-Hamiltonian and 3-regular. *Non-Hamiltonian* means that there is no closed 28-route visiting all the vertices; 3-*regular* means that the automorphism group is transitive on the open 3-routes (such as AB, BC, CD) but not on the 4-routes (such as AB, BC, CD, DE). Since the order of the automorphism group of an s-regular trivalent graph is 2^s times the number of edges [**26**, p. 71; **6**, p. 111], the group in the present case must have order $2^3.42 = 336$, which makes it precisely PGL(2, 7).

This 'Coxeter graph' was discovered independently by J. H. Conway and R. M. Foster. It was studied in detail by Biggs [**1**], who proved that it shares with the

Petersen graph, but with no other trivalent graph, the following three properties: its automorphism group acts primitively on the vertices and transitively on the pairs of vertices at each particular distance apart, and it is non-Hamiltonian.

A referee has pointed out that, like the Petersen graph, the Coxeter graph 'only just' fails to be Hamiltonian: although it admits no closed route visiting all the vertices, there is an open route doing so, for instance,

03 5∞ 12 06 45 1∞ 36 04 2∞ 56 34 0∞ 16 35
02 46 13 05 24 3∞ 15 26 4∞ 01 23 6∞ 14 25.

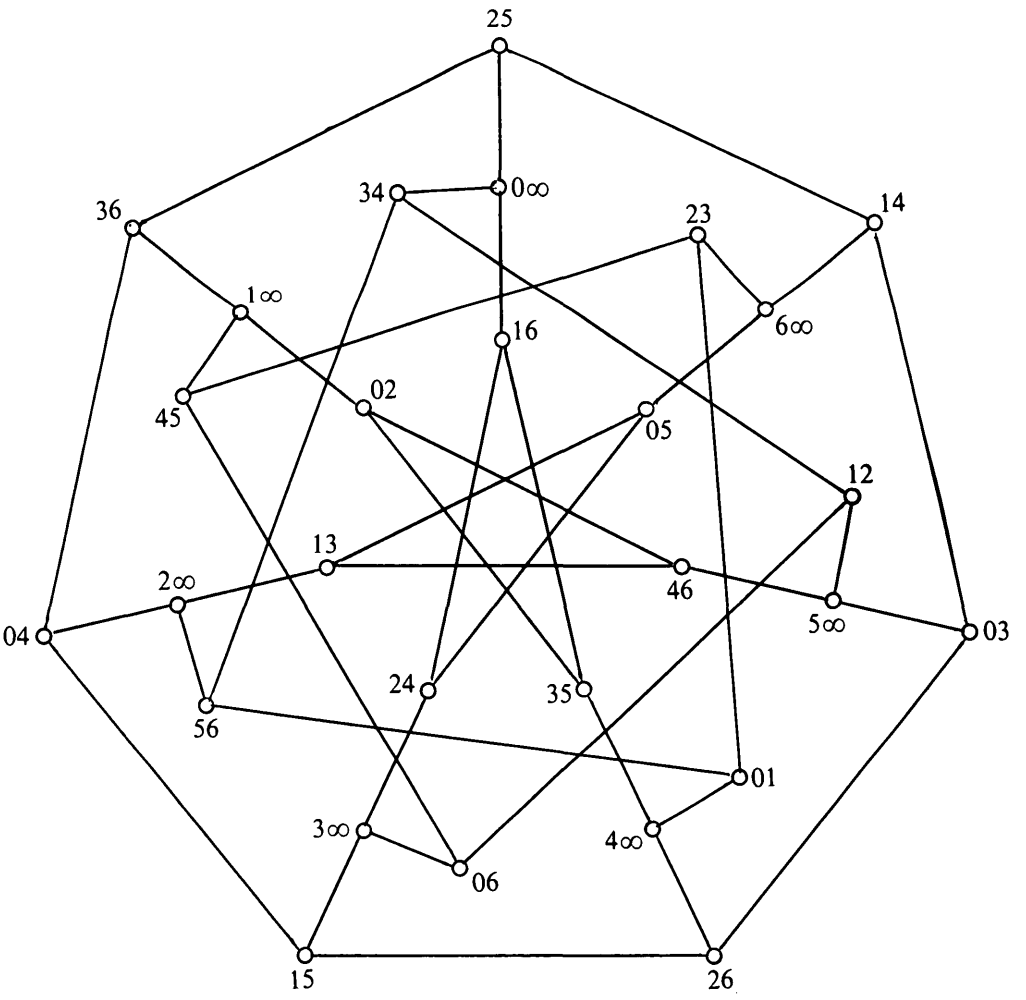


FIG. 2. A new notation for the Coxeter graph.

3. *Regular maps*

If we regard the right-hand half of Fig. 1 as a perspective view of the regular dodecahedron $\{5, 3\}$ (pentagonal faces, three at each vertex), the peripheral decagon represents a skew decagon of the kind that is called a *Petrie polygon*: every two adjacent edges belong to one face, but the next edge of the skew polygon belongs to a

different face [**10**, pp. 53, 111–112]. Accordingly it is convenient to amplify the Schläfli symbol, making it $\{5,3\}_{10}$, to indicate the decagonal Petrie polygon. The numbering of the vertices indicates the derivation of the *hemi-dodecahedron* $\{5,3\}_5$ by identifying pairs of opposite vertices. The resulting map of six pentagons covering the elliptic (or projective) plane exhibits one of two possible ways to embed the Petersen graph in such a plane. The pentagon 01 23 04 12 34 is a face, and the pentagon 02 14 03 24 13 is a Petrie polygon. The second embedding is derived by doubling all the residues (mod 5), so that 02 14 03 24 13 is now a face and 04 23 01 34 12 (or 01 23 04 12 34) is a Petrie polygon. Thus the symmetry group of either map is a subgroup of index 2 in the automorphism group of the graph. In fact, the former is the icosahedral group $\mathfrak{A}_5$, whereas the latter is the symmetric group $\mathfrak{S}_5$.

The alternating group $\mathfrak{A}_5$ of order 60 (known as the icosahedral group because it is the rotation group of the icosahedron $\{3,5\}$ and dodecahedron $\{5,3\}$) is generated by the even permutations

$$A = (1\ 4\ 2), \quad B = (0\ 1\ 2\ 3\ 4), \quad C = (0\ 1\ 3\ 2\ 4),$$

which exhibit it as the member $G^{3,5,5}$ of the family of groups $G^{p,q,r}$:

$$A^p = B^q = C^r = (BC)^2 = (CA)^2 = (AB)^2 = (ABC)^2 = 1 \tag{3.1}$$

[**4**, pp. 104, 113, 114, 130; **10**, pp. 96, 112]. Such a group occurs for many (but not all) values of the integers p, q, r. When it does occur, it is the automorphism group of each of the six maps

$$\{p,q\}_r, \quad \{q,r\}_p, \quad \{r,p\}_q, \quad \{q,p\}_r, \quad \{r,q\}_p, \quad \{p,r\}_q.$$

For instance, in $\{q,p\}_r$, A permutes the p edges at a vertex, B rotates a q-gonal face, and C displaces a Petrie polygon one step along itself. (In geometric terms, A and B are rotations while C is a glide-reflection.)

The above relations are easily seen to imply that $A^2B^2C^2 = 1$ so that, when $p = 3$, we have $A = B^2C^2$, and the group is generated by B and C alone:

$$B^q = C^r = (BC)^2 = (B^2C^3)^2 = (B^3C^2)^2 = 1.$$

If also $q = r$, there is an automorphism D of period 2 which interchanges B and C, so that

$$D^2 = 1, \quad DBD = C, \quad DCD = B.$$

Thus $G^{3,q,q}$ is a subgroup of index 2 in a group generated by B and D, with the presentation

$$B^q = D^2 = (BD)^4 = (B^3DB^2D)^2 = 1. \tag{3.2}$$

When $q = 5$, this is the automorphism group of the Petersen graph and of the Desargues configuration 10_3, that is, the symmetric group $\mathfrak{S}_5$ generated by the permutations

$$B = (0\ 1\ 2\ 3\ 4), \quad D = (2\ 3).$$

Similarly $G^{3,6,6}$, of order 108, is the automorphism group of the Pappus configuration 9_3 [**8**, p. 263]. It is a subgroup of index 2 in the group (3.2) with $q = 6$, which is the group of collineations *and correlations* of the same 9_3.

There is no $G^{3,7,7}$ [**24**, pp. 578–579], but $G^{3,8,8}$, of order 672, is the automorphism group $\mathfrak{C}_2 \times \mathrm{PGL}(2,7)$ of the map $\{8,3\}_8$ of 42 octagons on a surface of genus 8. It is a subgroup of index 2 in the group (3.2) with $q = 8$, which is the automorphism group of the graph formed by the 112 vertices and 168 edges of that map. This graph of girth 8 appears as $112B$ in the *Census* of Foster [**17**, p. 20]. Foster describes $112B$ as a 3-regular graph of girth 8_6 and diameter 10_1. 'Girth 8_6' means that the smallest circuit is an octagon and that each vertex belongs to six such circuits. 'Diameter 10_1' means that each vertex has a unique opposite, reached by a 10-route. Because of this uniqueness (indicated by the subscript 1), we may identify each pair of opposite vertices to obtain a graph $56C$ which is still 3-regular of girth 8_6 [**17**, p. 19]; but now the diameter is 7_1. Again the vertices occur in opposite pairs. Identifying them, we obtain 'graph 28' [**17**, p. 18] of girth 7, which is our friend the Coxeter graph (anticipated by Foster in his unpublished work of 1952).

Abraham Sinkov has found that, when $q = 8$, the relations (3.2) imply that $(B^3D)^7$, of period 4, is transformed into its inverse by B, and therefore also by D. Hence this group of order 1344 has a normal subgroup $\mathfrak{C}_4$, generated by $(B^3D)^7$. The quotient group is PGL(2, 7), with the new presentation

$$B^8 = D^2 = (BD)^4 = (B^3DB^2D)^2 = (B^3D)^7 = 1$$

and the representation

$$B = (0\ 1\ 4\ 2\ \infty\ 5\ 3\ 6), \quad D = (0\ 3)(2\ 5)(6\ \infty).$$

4. *The secants of a conic in* PG(2, 7)

The group PGL(2, 7) appears in geometry as the group of projectivities on the finite projective line PG(1, 7), or on a line or conic in the plane PG(2, 7). In order to relate the vertices of the Coxeter graph to the secants of such a conic, we seek a systematic way to associate the eight points on the conic

$$x^2 + y^2 + z^2 = 0 \tag{4.1}$$

with the elements $\alpha = 0, 1, 2, 3, 4, 5, 6, \infty$ of the extended GF[7]. Accordingly we consider the pencil of eight lines through the point (1, 2, 4) on the conic. One of them has the equation

$$x + y + z = 0;$$

another is the tangent $x + 2y + 4z = 0$. Taking linear combinations of these two, we use the element α to name the second intersection of the conic with the secant

$$x + y + z + \alpha(x + 2y + 4z) = 0,$$

and naturally $\alpha = \infty$ for the tangent. Thus the eight points on the conic are

0	1	2	3	4	5	6	∞
(2, 1, 4),	(1, 3, 2),	(3, 2, 1),	(2, 3, 1),	(2, 1, 3),	(1, 2, 3),	(3, 1, 2),	(1, 2, 4)

Finally, the pair $\alpha\beta$ provides a name for the secant joining points α and β; for instance, 01, joining (2, 1, 4) and (1, 3, 2), is the line [1, 0, 3], meaning $x + 3z = 0$. Thus the 28

secants are:

$0\infty = [1,1,1]$,	$01 = [1,0,3]$,	$02 = [0,3,1]$,	$03 = [1,2,-3]$,
$1\infty = [-1,2,1]$,	$12 = [-3,1,0]$,	$13 = [1,-1,1]$,	$14 = [0,-3,1]$,
$2\infty = [2,1,-1]$,	$23 = [1,1,2]$,	$24 = [1,0,-3]$,	$25 = [1,-2,1]$,
$3\infty = [3,0,1]$,	$34 = [-2,1,1]$,	$35 = [0,1,-3]$,	$36 = [1,-3,0]$,
$4\infty = [1,-1,2]$,	$45 = [1,1,-1]$,	$46 = [1,2,1]$,	$04 = [3,1,0]$,
$5\infty = [1,3,0]$,	$56 = [-3,0,1]$,	$05 = [-1,1,2]$,	$15 = [2,1,1]$,
$6\infty = [0,1,3]$,	$06 = [2,-1,1]$,	$16 = [1,1,-2]$,	$26 = [-1,1,1]$.

Edge [**12**, p. 365] explained why these twenty-eight are just those lines $[X, Y, Z]$ for which $X^2+Y^2+Z^2$ is a *nonsquare* in the field GF[7], namely 3 or 5 or 6. (As these 'line coordinates' are homogeneous, there are several ways to express them; for instance, $[1,0,3]$ could have been written as $[2,0,-1]$ or $[3,0,2]$. The above choice, in each case, has a reason that will appear in § 8.)

Two lines $[X, Y, Z]$ and $[X', Y', Z']$ are *conjugate* with respect to the conic $x^2+y^2+z^2=0$ (or $X^2+Y^2+Z^2=0$) if $XX'+YY'+ZZ'=0$. We can easily verify that every edge of the Coxeter graph represents a pair of conjugate secants (or their poles, which are conjugate external points). In other words, the 42 edges of the graph represent the 42 *harmonic tetrads* among the eight points of the projective line PG(1, 7) [**19**, pp. 120, 130].

5. *Unitary space*

The role of (3.1) as the automorphism group of the regular map $\{r,p\}_q$ may be elucidated by means of the substitution

$$T = AB, \quad R_2 = ABC, \quad R_3 = BC; \quad A = R_2R_3, \quad B = R_3R_2T, \quad C = TR_2.$$

This gives $G^{p,q,r}$ the alternative presentation

$$T^2 = R_2{}^2 = R_3{}^2 = (TR_2)^r = (R_2R_3)^p = (R_3T)^2 = (TR_2R_3)^q = 1.$$

Then T and R_2 generate the symmetry group $\mathfrak{D}_r$ of a face $\{r\}$ of $\{r,p\}_q$; T (reflecting in an 'inradius') interchanges the two ends of an edge ε, while R_2 (reflecting in a 'circumradius') interchanges ε with an adjacent edge. R_3 (reflecting in ε itself) interchanges the face $\{r\}$ with its neighbour [**10**, pp. 54, 111].

If q and r are even, say

$$q = 2m, \quad r = 2l,$$

the three reflections R_2, R_3 and $TR_2T = R_1$ (say) generate a subgroup of index 2:

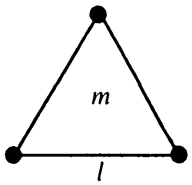
$$R_v{}^2 = (R_2R_3)^p = (R_3R_1)^p = (R_1R_2)^l = (R_1R_3R_2R_3)^m = 1.$$

This may be described as the subgroup of $G^{3,2l,2m}$ generated by A, BC, and CAB.

If, further, $p = 3$, the presentation

$$R_v{}^2 = (R_1R_2)^l = (R_1R_3R_2R_3)^m = 1, \quad R_2R_3R_2 = R_3R_2R_3, \quad R_3R_1R_3 = R_1R_3R_1$$

exhibits this as one of the *unitary reflection groups*, for which Shephard [**22**, p. 374] proposed a symbol closely resembling my



[**5**, pp. 247–248, 265]. The vertices of this triangular symbol represent the three generating reflections R_ν, with R_3 at the top. The m in the middle indicates the period of either of the conjugate elements $R_1R_3R_2R_3$, $R_1R_3R_1R_2$, or of any of the ten other expressions obtained by permuting the subscripts 1, 2, 3. As a convenient abbreviation for the triangular symbol, we write

$$[1\ 1\ 1^l]^m.$$

Making our final specialization $l = m = 4$, we observe that the representation of degree 10,

$$\begin{aligned} A &= (0\ 6\ 5)(1\ \infty\ 4), \\ B &= (0\ 1\ 4\ 2\ \infty\ 5\ 3\ 6), \\ C &= (0\ \infty\ 3\ 1\ 4\ 5\ 6\ 2)(7\ 8), \end{aligned}$$

for the group $G^{3,8,8} \simeq \mathfrak{C}_2 \times \mathrm{PGL}(2,7)$, yields, for the subgroup

$$[1\ 1\ 1^4]^4 = \mathfrak{C}_2 \times \mathrm{PSL}(2,7), \tag{5.1}$$

the representation

$$\begin{aligned} R_1 &= CAB = (0\ 2)(1\ 4)(3\ 5)(6\ \infty)(7\ 8), \\ R_2 &= ABC = (1\ 6)(2\ 3)(4\ 5)(0\ \infty)(7\ 8), \\ R_3 &= BC = (0\ 4)(1\ 5)(2\ 3)(6\ \infty)(7\ 8). \end{aligned}$$

Since

$$R_1R_2R_3 = (0\ 2\ 6\ 4\ \infty\ 5\ 3)(7\ 8),$$

the central $\mathfrak{C}_2$ is generated by $(R_1R_2R_3)^7 = (7\ 8)$.

According to Shephard and Todd [**23**, p. 295], the generators R_ν and their conjugates are reflections (of period 2) in the 21 planes

$$\begin{aligned} x &= 0, \quad y \pm z = 0, \quad cx \pm y \pm z = 0, \\ y &= 0, \quad z \pm x = 0, \quad cy \pm z \pm x = 0, \\ z &= 0, \quad x \pm y = 0, \quad cz \pm x \pm y = 0, \end{aligned}$$

where $c = \frac{1}{2}(-1 + i\sqrt{7})$, so that c and $\bar{c}$ are the roots of the quadratic equation

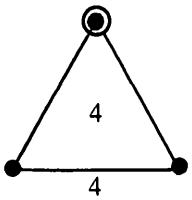
$$x^2 + x + 2 = 0. \tag{5.2}$$

The generators themselves, being reflections in the planes

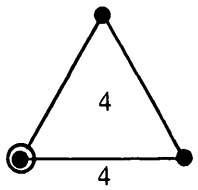
$$x = 0, \quad x = y, \quad x - cy + z = 0$$

[**5**, p. 260], are as follows: R_1 reverses the sign of x, R_2 interchanges x and y, R_3 takes (x, y, z) to $(\frac{1}{2}(x + cy - z), \frac{1}{2}\bar{c}(x + z), \frac{1}{2}(-x + cy + z))$.

Applying this group to the two points $(0, 0, 2)$ (on the first two mirrors) and $(1, 1, \bar{c}^2)$ (on the last two) in turn, we obtain two *complex polyhedra*



$(1\ 1\ 1_1{}^4)^4$ and



$(1_1\ 1\ 1^4)^4$

The polyhedron $(1\ 1\ 1_1{}^4)^4$ has 42 vertices, given by the permutations of

$$(\pm 2, 0, 0), \quad (0, \pm c, \pm c), \quad (\pm \bar{c}, \pm 1, \pm 1);$$

it has 168 edges and 112 triangular faces. In fact, it is a metric realization of the regular map $\{3, 8\}_8$, dual to the $\{8, 3\}_8$ described in § 3. The polyhedron $(1_1\ 1\ 1^4)^4$ has for its 56 vertices the centres of half these triangles; this too has 168 edges, and its faces consist of 56 triangles and 42 squares. Each vertex belongs to three triangles and three squares, arranged alternately [**5**, p. 266].

6. *The complex projective plane*

The 42 vertices of the complex polyhedron $(1\ 1\ 1_1{}^4)^4$ obviously occur in 21 pairs of opposites. In other words, the polyhedron has 21 *diameters* (lines through the origin). By making the coordinates homogeneous, we can interpret the 21 diameters as 21 points in a projective plane. These 21 points have for their projective coordinates the permutations of

$$(1, 0, 0), \quad (0, \pm 1, 1), \quad (\bar{c}, \pm 1, \pm 1).$$

Similarly, the 21 planes of symmetry, mirrors for the unitary group (5.1), yield 21 lines

$$[1, 0, 0], \quad [0, \pm 1, 1], \quad [c, \pm 1, \pm 1],$$

which are the 'Hermitian polars' of the 21 points; and the unitary group is reduced to its simple factor group PSL(2, 7), of order 168, with the presentation

$$R_\nu{}^2 = (R_2R_3)^3 = (R_3R_1)^3 = (R_1R_2)^4 = (R_1R_2R_3R_2)^4 = (R_1R_2R_3)^7 = 1$$

and the representation

$$\begin{aligned} R_1 &= (0\ 2)(1\ 4)(3\ 5)(6\ \infty), \\ R_2 &= (1\ 6)(2\ 3)(4\ 5)(0\ \infty), \\ R_3 &= (0\ 4)(1\ 5)(2\ 3)(6\ \infty). \end{aligned}$$

As in § 5, R_1 reverses the sign of x, R_2 interchanges x and y, and R_3 takes (x, y, z) to $(x + cy - z, \bar{c}x + \bar{c}z, -x + cy + z)$.

Each point and the corresponding line are the centre and axis for one of the 21 harmonic homologies considered by Klein and Fricke [**20**, p. 709], except that we are using a different triangle of reference, in terms of which their quartic curve $z_1{}^3z_4 + z_4{}^3z_2 + z_2{}^3z_1 = 0$ becomes

$$x^4 + y^4 + z^4 + 3\bar{c}(y^2z^2 + z^2x^2 + x^2y^2) = 0 \tag{6.1}$$

[**5**, p. 260]. This form of the equation was discovered by Ciani [**3**, pp. 364–365]; see also [**15**, pp. 332, 338; **14**].

It was observed by Klein and Fricke [**20**, p. 712] that the centres and axes for the 21 harmonic homologies form a self-dual configuration 21_4. For instance, since

$$c\bar{c} = 2 \quad \text{and} \quad c+\bar{c}+1 = 0$$

(see (5.2)), the line $[c, 1, 1]$ passes through the four points

$$(-\bar{c}, 1, 1), \quad (1, \bar{c}, 1), \quad (1, 1, \bar{c}), \quad (0, 1, -1).$$

The harmonic homologies R_1, R_2, R_3 generate the group PSL(2, 7) of 168 projective collineations. But the configuration and the curve are invariant also for 168 *antiprojective* collineations, one of which, say T, takes (x, y, z) to $(-\bar{x}+\bar{y}, \bar{x}+\bar{y}, \bar{c}\bar{z})$, and $[X, Y, Z]$ to $[-\bar{X}+\bar{Y}, \bar{X}+\bar{Y}, c\bar{Z}]$. This new collineation arises from the symmetrical nature of the graphical symbol for the polyhedron $(1\ 1\ 1_1{}^4)^4$, that is, from the fact that the group PSL(2, 7) has an automorphism T interchanging R_1 and R_2:

$$TR_1T = R_2, \quad TR_2T = R_1, \quad TR_3T = R_3, \quad T^2 = 1.$$

In terms of the symbols 0, 1, 2, 3, 4, 5, 6, ∞, T is an odd permutation, namely

$$T = (1\ 5)(3\ 6)(2\ \infty); \tag{6.2}$$

therefore the enlarged group, of order 336, is PGL(2, 7). This is the complete group of collineations, projective and antiprojective, transforming the configuration and quartic curve into themselves.

Notice that, in applying T to an *equation*, we first change the coefficients to their complex conjugates and then replace x, y, z by $-x+y, x+y, cz$. For instance, the equation

$$Xx + Yy + Zz = 0$$

for the line $[X, Y, Z]$ becomes $\bar{X}(-x+y) + \bar{Y}(x+y) + \bar{Z}cz = 0$ or

$$(-\bar{X}+\bar{Y})x + (\bar{X}+\bar{Y})y + c\bar{Z}z = 0,$$

and the left-hand side of (6.1) becomes

$$\begin{aligned}(x-y)^4 + (x+y)^4 + (cz)^4 + 3c\{(x+y)^2(cz)^2 + (cz)^2(x-y)^2 + (x^2-y^2)^2\}\\ = (2+3c)(x^4+y^4) + c^4z^4 + 6c^3(y^2z^2+z^2x^2) + 6(2-c)x^2y^2\\ = c^4\{x^4+y^4+z^4+3\bar{c}(y^2z^2+z^2x^2+x^2y^2)\}.\end{aligned}$$

If we include correlations as well as collineations, the group of our configuration is

$$\mathfrak{C}_2 \times \mathrm{PGL}(2, 7) \cong G^{3.8.8}$$

of order 672, where the $\mathfrak{C}_2$ is generated by the Hermitian polarity

$$x' = \bar{X}, \quad y' = \bar{Y}, \quad z' = \bar{Z}; \quad X' = \bar{x}, \quad Y' = \bar{y}, \quad Z' = \bar{z},$$

which interchanges the point (x, y, z) with the line $[\bar{x}, \bar{y}, \bar{z}]$.

7. *The fourteen self-polar triangles*

With respect to the Hermitian polarity, there are two ways to distribute the 21 centres and 21 axes into seven *self-polar triangles*. In such a triangle, each vertex is the Hermitian pole of the opposite side. One obvious instance is the triangle of reference $(1, 0, 0)(0, 1, 0)(0, 0, 1)$; another is $(\bar{c}, 1, 1)(1, \bar{c}, 1)(1, 1, \bar{c})$, with sides

$[c, 1, 1][1, c, 1][1, 1, c]$. Fig. 3 provides a convenient diagram in which the triangles of the two types are distinguished by their orientation: $\triangle$ or ∇. In this arrangement, the fourteen triangles, along with seven hexagons, form a quasi-regular map $\left\{\begin{matrix}3\\6\end{matrix}\right\}_{2,1}$ on a torus [**6**, pp. 115–118]. This symbol can be justified as follows. We begin with

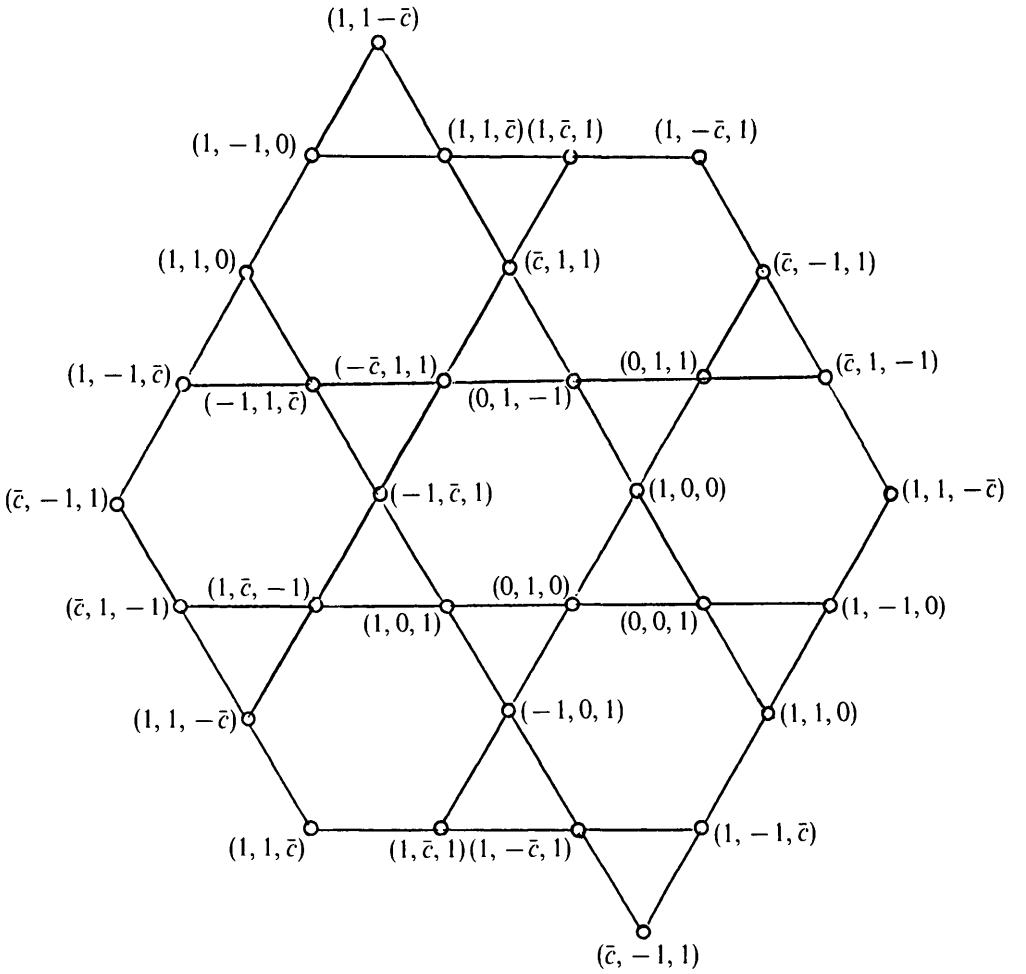


FIG. 3. The fourteen self-polar triangles exhibited as a quasi-regular map $\left\{\begin{matrix}3\\6\end{matrix}\right\}_{2,1}$, one of eight ways to embed the edge graph of the Heawood graph on the torus.

Heawood's regular map $\{6, 3\}_{2,1}$ (Fig. 4, which one uses to show that the colouring of a map on a torus may need as many as seven colours, because each of the seven hexagons is adjacent to all the remaining six). Converting the horizontal Schläfli symbol $\{6, 3\}$ to its vertical variant $\left\{\begin{matrix}3\\6\end{matrix}\right\}$ means taking as vertices the 'midpoints' of the 21 edges of $\{6, 3\}$ and joining pairs of them so as to make a triangle round each vertex of $\{6, 3\}$ and a hexagon inscribed in each face. The subscript $\{\ \}_{2,1}$ indicates the oblique 'knight's move' that relates pairs of peripheral points needing to be identified in order to change the infinite Euclidean plane into the finite torus. As this map is

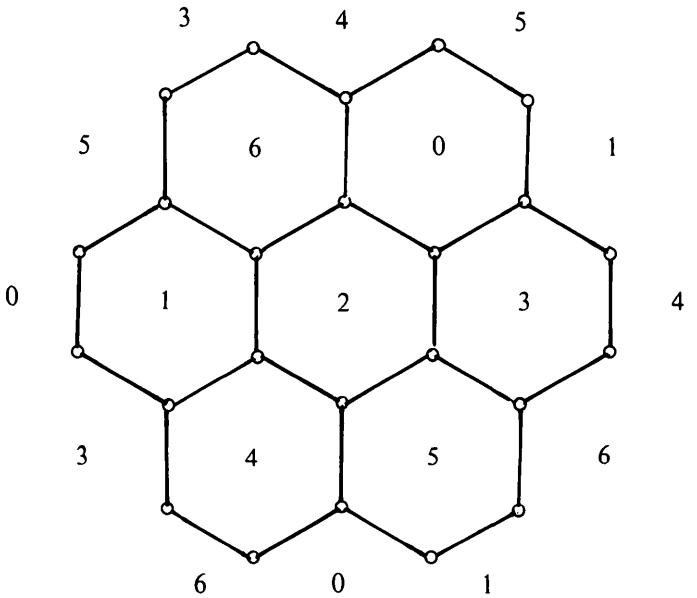


FIG. 4. The Heawood map $\{6,3\}_{2,1}$ on the torus.

chiral and has seven hexagonal faces, its automorphism group has order 42. In fact, this is the *metacyclic* subgroup of PGL(2, 7) which allows the residues modulo 7 to be increased by 1 (or more) or multiplied by 3 (or anything else), but not to be reciprocated [**10**, p. 11]. (Elsewhere it has been named $G^{3,2,-2}$ [**9**, p. 143].) But the tetravalent *graph* formed by the vertices and edges (the *edge graph* of the Heawood graph) has the group PGL(2, 7), of order 336. Hence Fig. 3 is one of *eight* ways to embed the graph on a torus. In fact, keeping the triangle $(1,0,0)(0,1,0)(0,0,1)$ fixed, we can flip over each of the three neighbouring triangles independently, and still complete a map $\left\{\begin{matrix}3\\6\end{matrix}\right\}_{2,1}$.

Figure 5 is a 'quasi-real' drawing in which the points $(0,1,-1)$, $(-1,0,1)$, $(1,-1,0)$ have been projected to infinity. The unreality of c necessitates certain anomalies such as the double occurrence of lines $[0,1,1]$ and $[-c,1,1]$. The 21 homology axes, which pass by fours through the 21 homology centres, also pass by threes through 28 new points, arising from the 28 diameters of either of the two congruent quasi-regular polyhedra $(1_1\ 1\ 1^4)^4$, $(1\ 1_1\ 1^4)^4$. One of these 28 new points can be seen in the middle of Fig. 5. From this point $(1,1,1)$, by applying the group generated by R_1, R_2, R_3, we can derive the whole set of 28: the permutations of

$$(1,\pm1,\pm1),\quad (\bar{c}^2,\pm1,\pm1),\quad (c,\pm1,0). \tag{7.1}$$

Using these points, we replace the self-dual configuration $21_4 = 21_4 21_4$ by $28_3 21_4$: a notation in which the complete quadrangle and complete quadrilateral would be named $4_3 6_2$ and $6_2 4_3$. Applying the Hermitian polarity to $28_3 21_4$, we obtain the dual configuration $21_4 28_3$, involving the 28 lines

$$[1,\pm1,\pm1],\quad [c^2,\pm1,\pm1],\quad [\bar{c},\pm1,0]. \tag{7.2}$$

Figure 5 reveals the fact that the two self-polar triangles

$$(1,0,0)(0,1,0)(0,0,1) \quad\text{and}\quad (\bar{c},1,1)(1,\bar{c},1)(1,1,\bar{c})$$

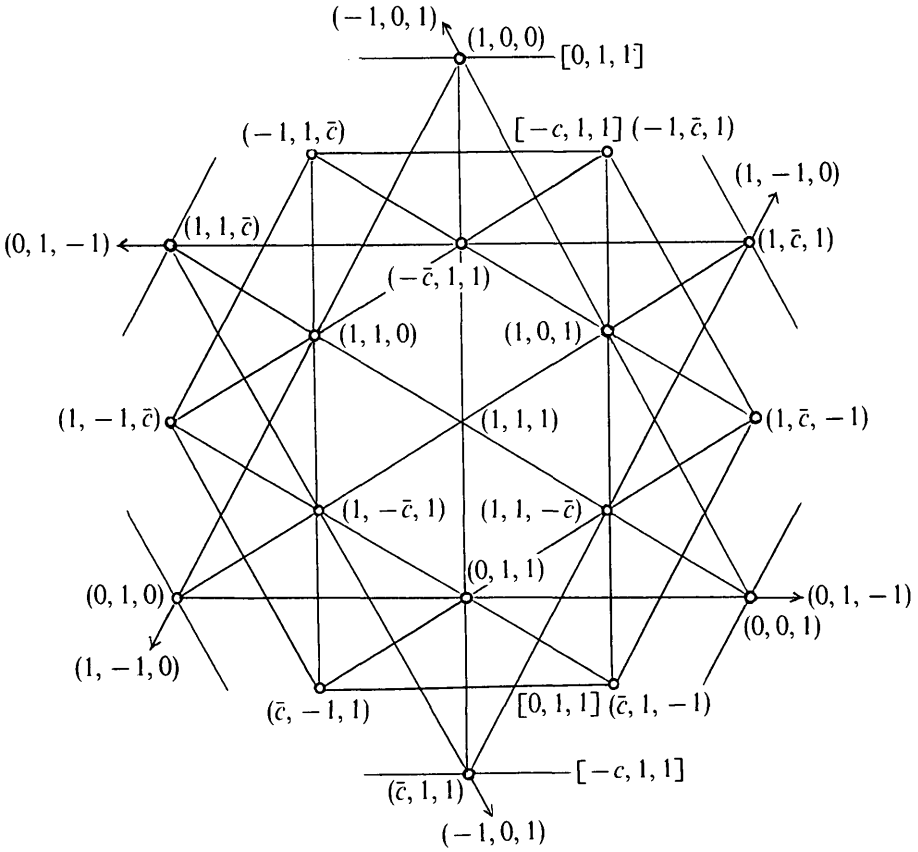


FIG. 5. The configuration 21_4.

are perspective triangles with centre of perspective $(1, 1, 1)$ and axis $[1, 1, 1]$. Hence the complete configuration, of $21+28$ points and $21+28$ lines, admits 28 subconfigurations, each of which is a Desargues 10_3. Comparison with Fig. 6 of an earlier paper [7, p. 239] shows that this is the kind of Desargues configuration whose collineation group is $\mathfrak{D}_6$, the dihedral group of order 12.

Although each of the 28 points lies on three of the 21 lines, and each of the 28 lines contains three of the 21 points, there are no incidences among the 28 points and the 28 lines; for instance, the equation $x+y+z=0$ is not satisfied by any of the points (7.1). On the other hand, since

$$\binom{21}{2} = 28\binom{3}{2} + 21\binom{4}{2},$$

the $28+21$ points account for *all* the intersections of the 21 lines with one another and, dually, the $28+21$ lines account for all the joins of pairs of the 21 points.

8. *The twenty-eight bitangents of the quartic curve*

Since the general plane quartic has 28 bitangents, it is natural to guess that the 28 lines (7.2) may be the bitangents of the curve (6.1). Since the group PGL(2, 7) permutes the lines and preserves the curve, it will suffice to consider just one of the

lines, such as $[-\bar{c}, 1, 0]$ or $y = \bar{c}x$. The intersections of this line with the curve can be found by substituting $\bar{c}x$ for y in the equation (6.1):

$$z^4 + 3\bar{c}(\bar{c}^2+1)z^2x^2 + (\bar{c}^4 + 3\bar{c}^3 + 1)x^4 = 0.$$

Since $\bar{c}$ satisfies the equation (5.2), we have

$$3\bar{c}(\bar{c}^2+1) = 3\bar{c}(-\bar{c}-1) = 3(\bar{c}+2-\bar{c}) = 6,$$
$$\bar{c}^4 + 3\bar{c}^3 + 1 = 3\bar{c}+2+3(2-\bar{c})+1 = 9.$$

Thus the left-hand side of the equation for z/x is simply

$$z^4 + 6z^2x^2 + 9x^4 = (z^2+3x^2)^2.$$

Since this is a perfect square, the four points of intersection coincide in pairs. Since our quartic curve has no double points, we conclude that the line $[-\bar{c}, 1, 0]$ is indeed a bitangent, its two points of contact with the curve being $(1, \bar{c}, \pm i\sqrt{3})$. Hence *all* the 28 lines (7.2) are bitangents.

Klein and Fricke [**20**, p. 707] reached the same conclusion from their different point of view.

We now see clearly how this quartic curve differs from the most general quartic curve, which likewise has 28 bitangents: in the present case there are 21 sets of four *concurrent* bitangents.

To find how to name the $\binom{8}{2}$ bitangents by pairs of the symbols $0, 1, \ldots, 6, \infty$, we pass from the field of complex numbers to the Galois field GF[7] by reducing all the integers to their residues modulo 7. Since

$$x^2 + x + 2 = (x-3)^2$$

in this field, we must replace c and $\bar{c}$ by 3, and c^2 and $\bar{c}^2$ by 2. The invariant quartic form

$$f = x^4 + y^4 + z^4 + 3\bar{c}(y^2z^2 + z^2x^2 + x^2y^2) \tag{8.1}$$

(see (6.1)) reduces to $(x^2+y^2+z^2)^2$. The configuration $21_4 21_4$ (Fig. 5) now consists of the 21 internal points and 21 nonsecants of this conic, namely the points (x, y, z) and lines $[x, y, z]$ for which $x^2+y^2+z^2$ is a non-zero square. Similarly, $28_3 21_4$ consists of the same 21 nonsecants along with the 28 external points (x, y, z) for which $x^2+y^2+z^2$ is a nonsquare; and $21_4 28_3$ consists of the 21 internal points and 28 secants. Now we have the pleasant surprise of a fourth configuration

$$28_3 28_3$$

formed by the 28 external points and 28 secants; for instance, the secant $[1, 1, 1]$ passes through the three external points

$$(-2, 1, 1), \quad (1, -2, 1), \quad (1, 1, -2),$$

even though the analogous points $(-\bar{c}^2, 1, 1)$, $(1, -\bar{c}^2, 1)$, $(1, 1, -\bar{c}^2)$ in the complex plane are *not* collinear.

Conversely, we can pass from the table of 28 secants in § 4 to an analogous table of bitangents by writing $\bar{c}$ instead of 3, and c^2 instead of 2.

Thus the 28 vertices of the Coxeter graph (Fig. 2) represent the 28 bitangents (or their poles) in such a way that joined vertices represent Hermitian conjugate lines (or points).

9. *Twenty-one conics from quadruples of concurrent bitangents*

Among the 28 bitangents of any non-singular quartic curve, a quadruple whose eight points of contact lie on a conic occurs 315 times [**21**, pp. 198–199]. In the case of the special quartic discussed by Klein and Fricke, 21 of the 315 conics arise from the 21 quadruples of *concurrent* bitangents. For instance, the homology centre (0, 0, 1) lies on the four bitangents

$$[\mp\bar{c}, 1, 0], \quad [1, \mp\bar{c}, 0], \tag{9.1}$$

whose eight points of contact are

$$(1, \pm\bar{c}, \pm i\sqrt{3}), \quad (\pm\bar{c}, 1, \pm i\sqrt{3}) \tag{9.2}$$

(with all combinations of sign), and these lie on the conic

$$3x^2+3y^2+cz^2 = 0. \tag{9.3}$$

The same result could have been attained, without computing the points of contact themselves, by considering the pencil of quartics $Q+\kappa f = 0$, where $Q = 0$ is the quadruple of bitangents and f is given by (8.1). A suitable value of κ yields a repeated conic [**20**, p. 714]; in fact,

$$(2-3\bar{c})(-\bar{c}x+y)(\bar{c}x+y)(x-\bar{c}y)(x+\bar{c}y)+c^2f = (3x^2+3y^2+cz^2)^2.$$

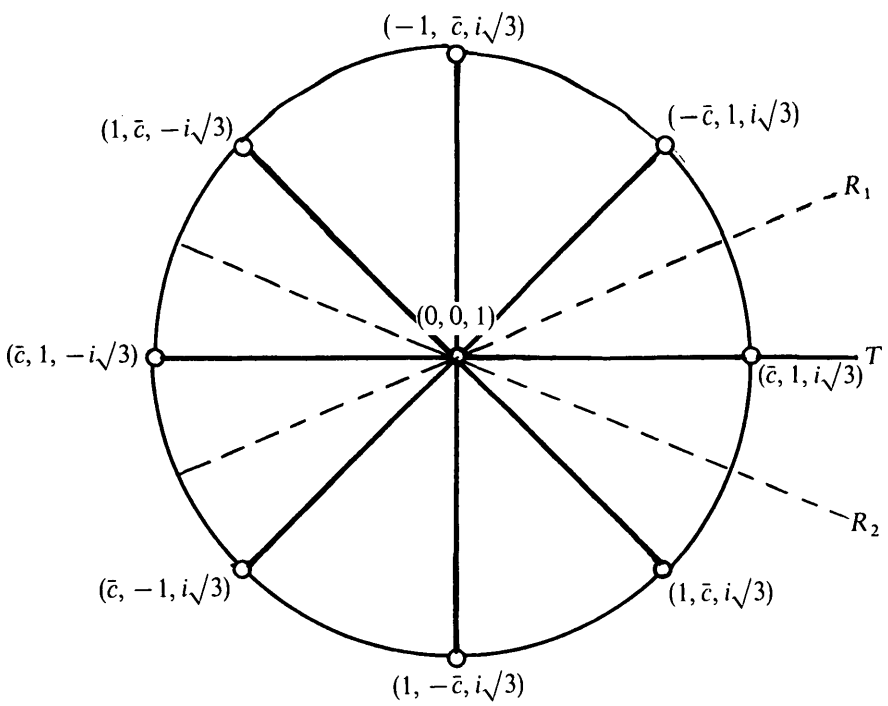
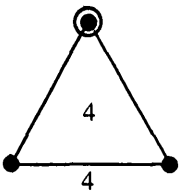
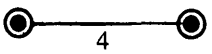


FIG. 6. The conic $3x^2+3y^2+cz^2 = 0$, drawn as a circle.

By the general rule for finding how a vertex of a uniform polytope is surrounded [**6**, p. 50], the vertex figure of



is



that is, the vertex figure of $(1\ 1\ 1_1{}^4)^4$ (see § 5) is a regular octagon, in agreement with the fact that the stabilizer of the point (0, 0, 1) in PGL(2, 7) is the dihedral group $\mathfrak{D}_8$ generated by R_1 and T (or by T and $R_2 = TR_1T$; see (6.2)). Fig. 6 exhibits the points of contact (9.2) as the vertices of a regular octagon cut out from the conic (9.3) by the four bitangents (9.1), one of which may be regarded as the mirror for the 'reflection' T which takes (x, y, z) to $(-\bar{x}+\bar{y}, \bar{x}+\bar{y}, \bar{c}\bar{z})$. (Notice how R_1 reverses the sign of x, or of both y and z, while R_2 interchanges x and y.)

In the notation of § 4, adapted as in § 8, the four bitangents (9.1) are

$$04,\ 12,\ 36,\ 5\infty.$$

In this cyclic order, each is transformed into the next by the element

$$R_2T = TR_1 = (0\ 2\ 6\ 5\ 4\ 1\ 3\ \infty)$$

of the $\mathfrak{D}_8$.

Applying the group generated by R_1, R_2, R_3, we can derive the whole set of 21 quadruples, and the corresponding conics, as in Table 1, where, to save space, we use

TABLE 1

Quadruple of bitangents	Point of concurrence	Conic
$02,\ 14,\ 35,\ 6\infty$	$(1, 0, 0)$	$cx^2+3y^2+3z^2=0$
$01,\ 24,\ 56,\ 3\infty$	$(0, 1, 0)$	$3x^2+cy^2+3z^2=0$
$04,\ 12,\ 36,\ 5\infty$	$(0, 0, 1)$	$3x^2+3y^2+cz^2=0$
$06,\ 13,\ 45,\ 2\infty$	$(0, 1, 1)$	$3x^2+cy^2+cz^2+b\bar{c}yz=0$
$03,\ 45,\ 26,\ 1\infty$	$(1, 0, 1)$	$cx^2+3y^2+cz^2+b\bar{c}zx=0$
$05,\ 26,\ 13,\ 4\infty$	$(1, 1, 0)$	$cx^2+cy^2+3z^2+b\bar{c}xy=0$
$15,\ 34,\ 26,\ 0\infty$	$(0, 1, -1)$	$3x^2+cy^2+cz^2-b\bar{c}yz=0$
$46,\ 25,\ 13,\ 0\infty$	$(-1, 0, 1)$	$cx^2+3y^2+cz^2-b\bar{c}zx=0$
$23,\ 16,\ 45,\ 0\infty$	$(1, -1, 0)$	$cx^2+cy^2+3z^2-b\bar{c}xy=0$
$05,\ 36,\ 24,\ 1\infty$	$(\bar{c}, 1, 1)$	$\bar{c}x^2+cy^2+cz^2+b\bar{c}yz-bzx-bxy=0$
$06,\ 35,\ 12,\ 4\infty$	$(1, \bar{c}, 1)$	$cx^2+\bar{c}y^2+cz^2-byz+b\bar{c}zx-bxy=0$
$03,\ 56,\ 14,\ 2\infty$	$(1, 1, \bar{c})$	$cx^2+cy^2+\bar{c}z^2-byz-bzx+b\bar{c}xy=0$
$01,\ 46,\ 23,\ 5\infty$	$(-\bar{c}, 1, 1)$	$\bar{c}x^2+cy^2+cz^2+b\bar{c}yz+bzx+bxy=0$
$04,\ 23,\ 15,\ 6\infty$	$(1, -\bar{c}, 1)$	$cx^2+\bar{c}y^2+cz^2+byz+b\bar{c}zx+bxy=0$
$02,\ 15,\ 46,\ 3\infty$	$(1, 1, -\bar{c})$	$cx^2+cy^2+\bar{c}z^2+byz+bzx+b\bar{c}xy=0$
$25,\ 01,\ 36,\ 4\infty$	$(\bar{c}, 1, -1)$	$\bar{c}x^2+cy^2+cz^2-b\bar{c}yz+bzx-bxy=0$
$16,\ 04,\ 35,\ 2\infty$	$(-1, \bar{c}, 1)$	$cx^2+\bar{c}y^2+cz^2-byz-b\bar{c}zx+bxy=0$
$34,\ 02,\ 56,\ 1\infty$	$(1, -1, \bar{c})$	$cx^2+cy^2+\bar{c}z^2+byz-bzx-b\bar{c}xy=0$
$24,\ 03,\ 16,\ 5\infty$	$(\bar{c}, -1, 1)$	$\bar{c}x^2+cy^2+cz^2-b\bar{c}yz-bzx+bxy=0$
$12,\ 05,\ 34,\ 6\infty$	$(1, \bar{c}, -1)$	$cx^2+\bar{c}y^2+cz^2+byz-b\bar{c}zx-bxy=0$
$14,\ 06,\ 25,\ 3\infty$	$(-1, 1, \bar{c})$	$cx^2+cy^2+\bar{c}z^2-byz+bzx-b\bar{c}xy=0$

the abbreviation

$$b = c - \bar{c} = i\sqrt{7}$$

(so that $bc = \bar{c} - 3$ and $b\bar{c} = 3 - c$).

The entries in this table have been grouped in sets of three corresponding to cyclic permutation of the three coordinates. In the first column, this permutation appears as (1 4 2)(3 5 6). Notice that all the conics reduce to (4.1) when we set $c = \bar{c} = 3$ and $b = 0$.

The incidences in the configuration $21_4 28_3$ show that each bitangent $l = 0$ belongs to three of the 21 quadruples; for instance, the bitangent $0\infty = [1, 1, 1]$ belongs to the third set of three rows in our table. Such a set of three quadruples yields three conics

$$s_v = 0 \quad (v = 1, 2, 3),$$

all passing through two points on $l = 0$. Every two of these three conics have two other points of intersection, joined by a line, say

$$l_v = 0 \quad (v = 1, 2, 3).$$

What happens now is most vividly seen by taking the two common points of the three conics to be the isotropic points in the extended Euclidean plane, so that $l = 0$ is the line at infinity [**20**, p. 714]. Then the three conics are circles, the lines $l_v = 0$ are their radical axes, and what happens is that these three lines are concurrent, all passing through the radical centre.

Alternatively, adding the three equations

$$s_2 - s_3 = ll_1, \quad s_3 - s_1 = ll_2, \quad s_1 - s_2 = ll_3,$$

we obtain $l(l_1 + l_2 + l_3) = 0$, whence $l_1 + l_2 + l_3 = 0$.

It is significant that the point of concurrence of these three 'residual' lines turns out to be the *Hermitian pole* of the original bitangent $l = 0$. For a proof, it suffices to let this bitangent be $0\infty = [1, 1, 1]$. Then

$$s_1 = 3x^2 + cy^2 + cz^2 - b\bar{c}yz, \quad \text{etc.},$$

as in our third column, and their differences are numerical multiples of ll_1, ll_2, ll_3, namely

$$s_2 - s_3 = b\bar{c}(x + y + z)(y - z), \quad \text{etc.},$$

so that

$$l = x + y + z, \quad l_1 = y - z, \quad l_1 = z - x, \quad l_3 = x - y,$$

and the point of concurrence is (1, 1, 1).

This may reasonably be regarded as a 'construction' for the Hermitian pole of the original bitangent $l = 0$.

10. *Klein's 7 + 7 conics from symmetric quadruples*

In each of the 21 quadruples that we have been discussing, the four bitangents are permuted by a dihedral subgroup $\mathfrak{D}_4$ of PSL(2, 7), belonging to a subgroup $\mathfrak{D}_8$ of PGL(2, 7). Klein and Fricke [**20**, p. 712] were particularly interested in fourteen other quadruples, each permuted by an octahedral subgroup $\mathfrak{S}_4$ of PGL(2, 7). These arise

from the fourteen 'self-polar' triangles exhibited in our Fig. 3. One of these is the triangle of reference, which is the diagonal triangle of the complete quadrilateral formed by the four bitangents

$$x+y+z=0, \quad x-y-z=0, \quad -x+y-z=0, \quad -x-y+z=0$$

(or $[1, \pm 1, \pm 1]$, or $0\infty, 13, 26, 45$), whose eight points of contact lie on the conic $x^2+y^2+z^2=0$, in accordance with the identity

$$b(x-y-z)(-x+y-z)(-x-y+z)(x+y+z)+c^2 f = \bar{c}^2(x^2+y^2+z^2)^2.$$

Applying the group PSL(2, 7) generated by R_1, R_2, R_3, and then the antiprojective collineation T which yields the rest of PGL(2, 7), we obtain two families of seven quadruples, and the corresponding conics, as in Table 2.

Notice that each family of seven quadruples includes all the 28 bitangents. In other words, the 56 points of contact of the 28 bitangents are the 56 intersections of the quartic curve with seven conics, and this happens in two ways [**18**, pp. 712, 715]. Since each bitangent belongs to one quadruple of each family, we can represent the $7+7$ quadruples by the $7+7$ vertices of a tetravalent bipartite graph of girth 4, whose 28 edges represent the 28 bitangents. In Fig. 7, the vertices marked u_α and v_α represent the quadruples containing the bitangent $\alpha\infty$ ($\alpha = 0, 1, \ldots, 6$), and it is convenient to let the same symbols denote the left-hand sides of the equations of the corresponding conics. If $l=0$ is the common bitangent of two quadruples u_α and v_β (that is, the bitangent represented by the edge $u_\alpha v_\beta$ of the graph), we observe that l^2 is a linear combination of the quadratic forms u_α and v_β. For instance,

$$b(x+y+z)^2 = -\bar{c}^4 u_0 - c v_0,$$

$$b(x+\bar{c}z)^2 = \bar{c} u_3 + c^3 v_5,$$

$$b(c^2 x+y+z)^2 = c^3 u_6 - c^3 v_0,$$

Hence, *each bitangent is associated with two conics, one in each family of seven, and these two conics have double contact. The bitangent's Hermitian pole is its common pole with respect to the two conics.*

Figure 7 epitomizes not only Table 2 but also Table 1, in which each bitangent belongs to three of 21 quadruples. These three quadruples appear in the graph as three quadrangles with a common edge. Each edge determines three other edges, namely the opposite edge of each of the three quadrangles. For instance, the edge 0∞ yields the three 'parallel' edges 34, 25, 16, corresponding, in the Coxeter graph (Fig. 2), to the neighbouring vertices of the vertex 0∞.

In other words, each bitangent determines three other bitangents, opposite to it in the three quadruples to which it belongs (in Table 1). For instance, the bitangent $[1, 1, 1]$ yields

$$[-c^2, 1, 1], \quad [1, -c^2, 1], \quad [1, 1, -c^2].$$

Since

$$-bc(x+y+z)(-c^2x+y+z)(x-c^2y+z)(x+y-c^2z)+f$$
$$= \bar{c}^4\{\bar{c}(x^2+y^2+z^2)-b(yz+zx+xy)\}^2,$$

the quadruple 0∞, 34, 25, 16 consists of $1+3$ bitangents whose eight points of contact lie on the conic

$$\bar{c}(x^2+y^2+z^2)-b(yz+zx+xy)=0.$$

TABLE 2

Quadruple of bitangents	Coordinates	Conic
13, 26, 45, $0\varkappa$	$[1, \pm 1, \pm 1]$	$u_0 = x^2+y^2+z^2 = 0$
24, 03, 56, $1\varkappa$	$[-1, \pm c^2, 1], [1, 0, -\bar{c}], [-\bar{c}, 0, 1]$	$u_1 = x^2-c^3y^2+z^2-2bzx = 0$
35, 14, 06, $2\varkappa$	$[\pm c^2, 1, -1], [0, 1, -\bar{c}], [0, -\bar{c}, 1]$	$u_2 = -c^3x^2+y^2+z^2-2byz = 0$
46, 25, 01, $3\varkappa$	$[1, \pm c^2, 1], [1, 0, \bar{c}], [\bar{c}, 0, 1]$	$u_3 = x^2-c^3y^2+z^2+2bzx = 0$
05, 36, 12, $4\varkappa$	$[1, -1, \pm c^2], [1, -\bar{c}, 0], [-\bar{c}, 1, 0]$	$u_4 = x^2+y^2-c^3z^2-2bxy = 0$
16, 04, 23, $5\varkappa$	$[1, 1, \pm c^2], [1, \bar{c}, 0], [\bar{c}, 1, 0]$	$u_5 = x^2+y^2-c^3z^2+2bxy = 0$
02, 15, 34, $6\varkappa$	$[\pm c^2, 1, 1], [0, 1, \bar{c}], [0, \bar{c}, 1]$	$u_6 = -c^3x^2+y^2+z^2+2byz = 0$
15, 23, 46, $0\varkappa$	$[c^2, 1, 1], [1, c^2, 1], [1, 1, c^2], [1, 1, 1]$	$v_0 = x^2+y^2+z^2-b\bar{c}(yz+zx+xy) = 0$
26, 34, 05, $1\varkappa$	$[-c^2, 1, 1], [-1, c^2, 1], [-1, 1, c^2], [-1, 1, 1]$	$v_1 = x^2+y^2+z^2+b\bar{c}(-yz+zx+xy) = 0$
03, 45, 16, $2\varkappa$	$[c^2, 1, -1], [1, c^2, -1], [1, 1, -c^2], [1, 1, -1]$	$v_2 = x^2+y^2+z^2+b\bar{c}(yz+zx-xy) = 0$
14, 56, 02, $3\varkappa$	$[0, \pm\bar{c}, 1], [\pm\bar{c}, 0, 1]$	$v_3 = \bar{c}x^2+\bar{c}y^2+cz^2 = 0$
25, 06, 13, $4\varkappa$	$[c^2, -1, 1], [1, -c^2, 1], [1, -1, c^2], [1, -1, 1]$	$v_4 = x^2+y^2+z^2+b\bar{c}(yz-zx+xy) = 0$
36, 01, 24, $5\varkappa$	$[1, 0, \pm\bar{c}], [1, \pm\bar{c}, 0]$	$v_5 = cx^2+\bar{c}y^2+\bar{c}z^2 = 0$
04, 12, 35, $6\varkappa$	$[\pm\bar{c}, 1, 0], [0, 1, \pm\bar{c}]$	$v_6 = \bar{c}x^2+cy^2+\bar{c}z^2 = 0$

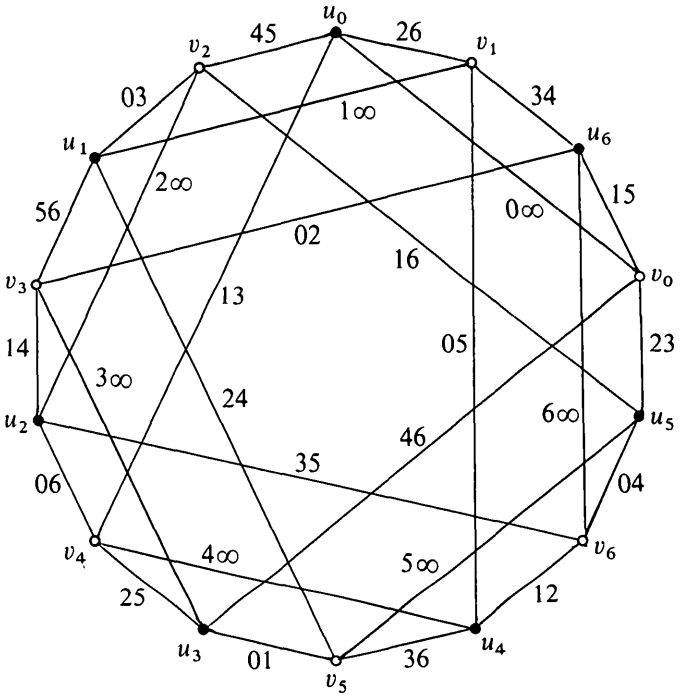


FIG. 7. The 7+7 quadruples of bitangents.

This is one of 28 conics (of the 'first kind', in the terminology of Klein and Fricke [**20**, p.714]). The equations of the remaining 27 may be left as an exercise for the reader.

In § 8 we passed from the field of complex numbers to the Galois field GF[7]. If we pass to GF[3^2] instead, we can identify c with a primitive element j of this field, letting the two moduli be 3 and j^2+j+2 $(= j^2+j-1)$. Then

$$j = c, \quad j^2 = b, \quad j^3 = \bar{c}, \quad j^4 = -1, \quad j^5 = -j = b\bar{c},$$

and the quartic form (8.1) reduces to $x^4+y^4+z^4$. When we substitute j, j^2, j^3, and $-j$ for c, b, $\bar{c}$, and $b\bar{c}$ in Table 2, our fourteen conics agree precisely with those displayed as symmetric matrices by Edge [**13**, p. 6]. In the same paper (on page 8) Edge warns us not to rely too heavily on Klein and Fricke [**20**, pp. 712–715], because an oversight of theirs caused four misprints.

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